LESSON 1.4

Math is Explaining and Sharing

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can construct an argument using equations, drawings, or words, and use volume to defend a recommendation.

WHY THIS MATTERS Yuzuki is buying her favorite breakfast cereal. She can buy three large-sized boxes or two jumbo-sized boxes for the same price. Which option should she pick — and can you convince a classmate you are right? Each large-sized box holds 240 cubic inches of cereal and each jumbo-sized box holds 400 cubic inches.

REMEMBER

Math is Explaining and Sharing

Math Practices: Mathematical Arguments & Justification Getting an answer is only half the work. The other half is explaining it so clearly — with conjectures, examples, and counterexamples — that someone else can decide whether your argument is convincing.

1. State a clear claim or conjecture supported by calculations.
2. Use accurate units and vocabulary (e.g. cubic units for volume).
3. Test with counterexamples to verify that the rule holds true every time

Words: argument *(argumento)* · conjecture *(conjetura)* · counterexample *(contraejemplo)* · volume *(volumen)* · cubic unit *(unidad cúbica)*

PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

Sort each question: area or volume?

Groups: Volume — cubic units · Area — square units

_____ How much soil fills the garden bed?

_____ How much cereal fits inside the box?

_____ How much water fills the fish tank?

_____ How much paper covers the lid of the box?

_____ How much paint covers one wall?

_____ How much mulch covers the top of the bed?

HINT 'Fill' means the inside of a 3-D shape — that is volume, in cubic units.

2 **Complete the table. Show how you found each value.**

Compute the volume of each cereal box.

Box (l × w × h, inches)	Volume (cubic inches)
8 × 3 × 10	
5 × 8 × 10	
6 × 4 × 5	

HINT Multiply two dimensions first, then multiply the result by the third.

SHOW YOUR WORK

PART 2 Practice the skill

show every step

3 **Circle the letter of the best answer. Show how you know.**

What is 2.5 × 10?

(A) 2.5
(B) 12.5
(C) 25
(D) 250

SHOW EACH PARTIAL PRODUCT, THEN ADD THEM

1. **Split** *Break one factor apart.*
2. **Multiply** *One part at a time.*
3. **Add** *Add the parts.*

PART 3 Apply it

real-world problems

4 **Circle the letter of the best answer. Show how you know.**

A tram carries 10 passengers each trip. How many trips move 40 passengers?

(A) 4 trips
(B) 10 trips
(C) 30 trips
(D) 400 trips

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

Multiplying a whole number by 10 makes the number…

(A) 10 more than it was
(B) ten times smaller
(C) exactly the same
(D) ten times as large

SHOW YOUR WORK

EXPLAIN YOUR THINKING

One of you sorted a card into 'Argument using equations' and the other into 'Argument using words.' Which argument would convince a skeptic faster, and what would it take to turn the other one into a counterexample?

SENTENCE STARTER An argument using ___ is stronger here because ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 1.4

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Set B · Version B · Second form · core practice

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- ☐ Meets target
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LEARNING TARGET I can construct an argument using equations, drawings, or words, and use volume to defend a recommendation.

WHY THIS MATTERS Yuzuki is buying her favorite breakfast cereal. She can buy three large-sized boxes or two jumbo-sized boxes for the same price. Which option should she pick — and can you convince a classmate you are right? Each large-sized box holds 240 cubic inches of cereal and each jumbo-sized box holds 400 cubic inches.

PART 1 Understand the idea

models and meaning

1 **Circle the letter of the best answer. Show how you know.**

A neighborhood group is building a community garden. To fill the beds they need soil — and to order the right amount, they need volume, measured precisely in cubic units. One design needs 48 cubic feet of soil. The group decides to decompose the garden bed design into simpler shapes to find its volume. What does that method involve?

(A) Breaking the figure into rectangular prisms, finding each prism's volume, then adding them

(B) Adding the areas of each face of the figure

(C) Multiplying the total length by the total width only

(D) Averaging the volumes of similar containers they've seen before

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

Which statement best captures what this lesson means by 'Math is Explaining and Sharing'?

(A) We keep our methods private so nobody can copy our answers

(B) An argument only counts in math if it uses equations

(C) We communicate our reasoning, listen to classmates' arguments, decide whether they are convincing, and check our calculations for accuracy

(D) Once you have an answer, checking it a second way is a waste of time

SHOW YOUR WORK

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

A neighborhood group is building a community garden. To fill the beds they need soil — and to order the right amount, they need volume, measured precisely in cubic units. One design needs 48 cubic feet of soil. The group needs to order soil to fill the garden beds. Which unit should they use to express the amount of soil?

(A) feet

(B) square feet

(C) cubic feet

(D) cubic inches

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

A neighborhood group is building a community garden. To fill the beds they need soil — and to order the right amount, they need volume, measured precisely in cubic units. One design needs 48 cubic feet of soil. After finding the volume of the garden beds one way, what should the group do to be confident their total of 48 cubic feet is accurate?

(A) Round the answer to the nearest ten

(B) Decompose the figure a second, different way and check that both totals match

(C) Multiply the total by 2 just to be safe

(D) Ask a classmate if the number sounds right

SHOW YOUR WORK

EXPLAIN YOUR THINKING

One of you sorted a card into 'Argument using equations' and the other into 'Argument using words.' Which argument would convince a skeptic faster, and what would it take to turn the other one into a counterexample?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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